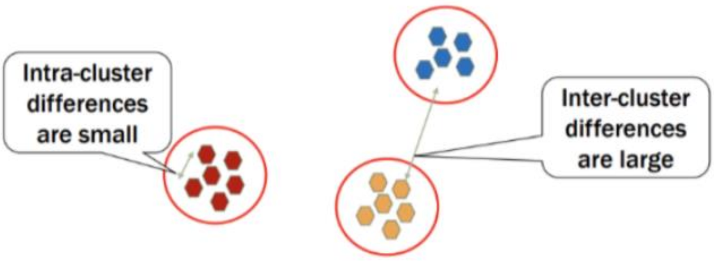
BMEG3105 Data analytics for personalized genomics and precision medicine Topic: Lecture8 Clustering

Scriber: 1155213539 HAN Mingyu

Definition of clustering analysis:

Finding groups of objects such that the objects in group

- will be similar to one another
- and different from the objects in other groups



Keypoint : Intra-cluster differences are small;Inter-cluster differences are large.

Minkowski distance

$$dist(\boldsymbol{p}, \boldsymbol{q}) = \left(\sum_{k=1}^{m} |p_k - q_k|^r\right)^{\frac{1}{r}}$$

r is a parameter;

r =1 .City block (Manhattan, taxicab, $L1$ norm) distance

This is the Hamming distance, which is just the number of bits.that are different between two binary vectors

r =2 .Euclidean distance

r ---> infinity (supremum distance)

- maximum distance difference between and component of vectors
-

Minkowski distance Examples

point	x	y
p1	0	2
p2	2	0
p3	3	1
p4	5	1

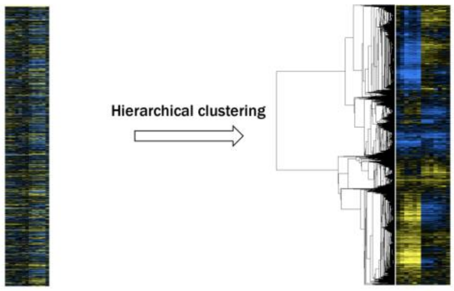
L1	p1	p2	p3	p4
p1	0	4	4	6
p2	4	0	2	4
p3	4	2	0	2
p4	6	4	2	0

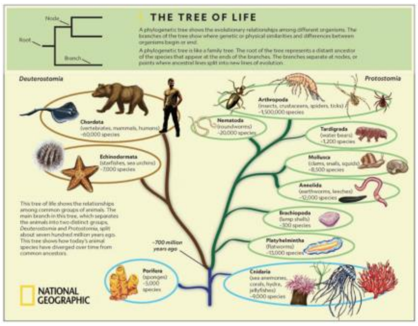
L2	p1	p2	p3	p4
p1	0	2.828	3.162	5.099
p2	2.828	0	1.414	3.162
p3	3.162	1.414	0	2
p4	5.099	3.162	2	0

L_∞	p1	p2	p3	p4
p1	0	2	3	5
p2	2	0	1	3
p3	3	1	0	2
p4	5	3	2	0

Hierarchical Clustering

- Definition: Produce a set of nested clusters organized as a hierarchical tree
- Can be visualized as a dendrogram, e.g. a tree diagram that records the sequences of merges
- They may correspond to meaningful taxonomies, e.g. e.g. gene clusters, phylogeny reconstruction, animal kingdom...





Schematic diagram

Steps of hierarchical clustering:

1. Compute the Similarity or Distance matrix
2. Let each data point be a cluster
3. Merge the two closest clusters
4. Update the similarity or distance matrix

   Ways to update the distance matrix after merging. E.g. min,max,group average,distance between centroids
5. Repeat the previous two steps...

In the hierarchical clustering, we can update the distance matrix without the original data matrix.

Running example:

Gene	wt	mutant_1	mutant_2	mutant_3
At4g35770	1.5	3	3	1.5
At1g30720	4	7.5	7.5	5
At4g27450	1.5	1	1	1.5
At2g34930	10	25	23	15
At2g05540	1	1	2	1

1. Compute distance matrix with linear correlation.

$$\rho_{X,Y} = \mathrm{corr}(X,Y) = \frac{\mathrm{cov}(X,Y)}{\sigma_X \sigma_Y} = \frac{\mathrm{E}[(X-\mu_X)(Y-\mu_Y)]}{\sigma_X \sigma_Y}$$

2. Each gene be a cluster.

	At4g35770	At1g30720	At4g27450	At2g34930	At2g05540
At4g35770	1				
At1g30720	0.9733	1			
At4g27450	-1	-0.9733	1		
At2g34930	0.9493	0.9909	-0.9493	1	
At2g05540	0.5774	0.562	-0.5774	0.4528	1

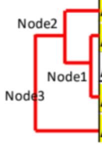
3.In the above table, The highest value besides 1 is 0.9909,we merge At2g34930 and At1g30720 and Remove 1.We take the highest correlation for At2g34930 and At1g30720.

	At4g35770	At1g30720	At4g27450	At2g34930	At2g05540
At4g35770					
At1g30720	0.9733				
At4g27450	-1	-0.9733 ->-0.9493			
At2g34930	0.9493 ->0.9733		-0.9493		
At2g05540	0.5774	0.562	-0.5774	0.4528 ->0.562	

4.Next highest value is 0.9733,we merge At4g35770 and At2g34930 , At1g30720 use the same method.

	At4g35770	At1g30720	At4g27450	At2g34930	At2g05540
At4g35770					
At1g30720					
At4g27450	-1 ->-0.9493	-0.9493			
At2g34930			-0.9493		
At2g05540	0.5774	0.562 ->0.5774	-0.5774	0.562 ->0.5774	

5. Next highest value is 0.5774 ,we merge At2g05540,At4g35770 and At2g34930 , At1g30720 use the same method.



	At4g35770	At1g30720	At4g27450	At2g34930	At2g05540
At4g35770					
At1g30720					
At4g27450	-0.5774	-0.5774			
At2g34930			-0.5774		
At2g05540			-0.5774		

This is the whole process：

1.Each gene be a cluster

2.Merge At2g34930 and At1g30720

3.Update with minimum distance (largest correlation)

4.Merge At2g34930, At1g30720 and At4g35770

5.Update with minimum distance (largest correlation)

6.Merge At2g34930, At1g30720, At4g35770, and At2g05540

Programming

Scikit-learn: https://scikit-learn.org/stable/

Hierarchical Clustering in python

https://scikit-learn.org/stable/auto_examples/cluster/plot_agglomerative_dendrogram

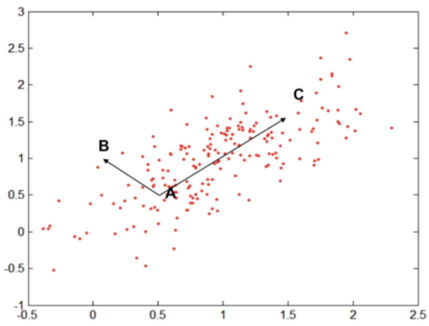
Protential project 2

Data preprocessing for the gene expression matrix

- Data collecting and merging (if needed)
- Exploration
- Visualization
- Data cleaning
- Get distance matrix
- Perform clustering

Dimension reduction (Lec-11)

Mahalanobis distance



- Calculating distance considering the data distribution

$$mahalanobis(p, q) = (p - q)^T \Sigma^{-1} (p - q)$$

Σ is the covariance matrix

How to calculate the inverse of the covariance matrix?

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \underbrace{\frac{1}{ad-bc}}_{\text{determinant}} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

$$\begin{bmatrix} 4 & 7 \\ 2 & 6 \end{bmatrix}^{-1} = \frac{1}{4\times6-7\times2} \begin{bmatrix} 6 & -7 \\ -2 & 4 \end{bmatrix} = \frac{1}{10} \begin{bmatrix} 6 & -7 \\ -2 & 4 \end{bmatrix} = \begin{bmatrix} 0.6 & -0.7 \\ -0.2 & 0.4 \end{bmatrix}$$

Example: Suppose we have two quizzes.

Quiz 1: std = 10

Student A – Student B = 1

Quiz 2: std = 1

Student A – Student B = 1

Should the two differences (A-B, C-D) be considered the same?

：The standard deviations of the two quizzes differ (10 points vs. 1 point), so a 1-point difference has entirely different statistical meanings. The gap between A and B is only 0.1 standard deviations, while the gap between C and D is 1 standard deviation, meaning the latter represents a much larger relative difference and thus should not be considered the same.

How to compare the two differences fairly?

: Compute how many standard deviations away between 2 points.